

loop statements

Example (1): Write C++ program using **while** loop to find the summation of the following series:

$$\text{sum} = 1 + 3 + 5 + 7 + \dots + 99$$

```
#include <iostream>
using namespace std;
int main() {
    int count = 1;
    int sum = 0;
    while ( count <= 99 )
    {
        sum = sum + count;
        count = count + 2;
    }
    cout << "sum is: " << sum << endl;

    return 0;
}
```

Output :

sum is: 2500

Example (2): Write C++ program using **for loop** to print the even numbers between 2 and 20:

```
#include <iostream>
using namespace std;
int main() {
    for (int i=2; i<=20; i+=2)
    {
        cout << i << endl;
    }
    return 0;
}
```

Output : 2

4
6
8
10
12
14
16
18
20

Laboratory task to groups A, B, C:

Write program using any statement loop to find the summation of the following series:

$$\sum_{i=1}^n i^n$$

Suppose $n=10$